

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 purchases** to guide strategic business decisions

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Project Overview



This analysis examines customer shopping behavior using transactional data from **3,900 purchases** across various product categories.

Goal: Uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to drive strategic business decisions.

3,900

Total Purchases

18

Data Columns

37

Missing Values

Data Preparation with Python

01

Data Loading & Exploration

Imported dataset using pandas, analyzed structure with `df.info()` and summary statistics

02

Column Standardization

Renamed columns to `snake_case` for better readability and documentation

03

Data Consistency

Verified redundancy between `discount_applied` and `promo_code_used`, dropped duplicate

04

Missing Data Handling

Imputed missing Review Rating values using median rating per product category

05

Feature Engineering

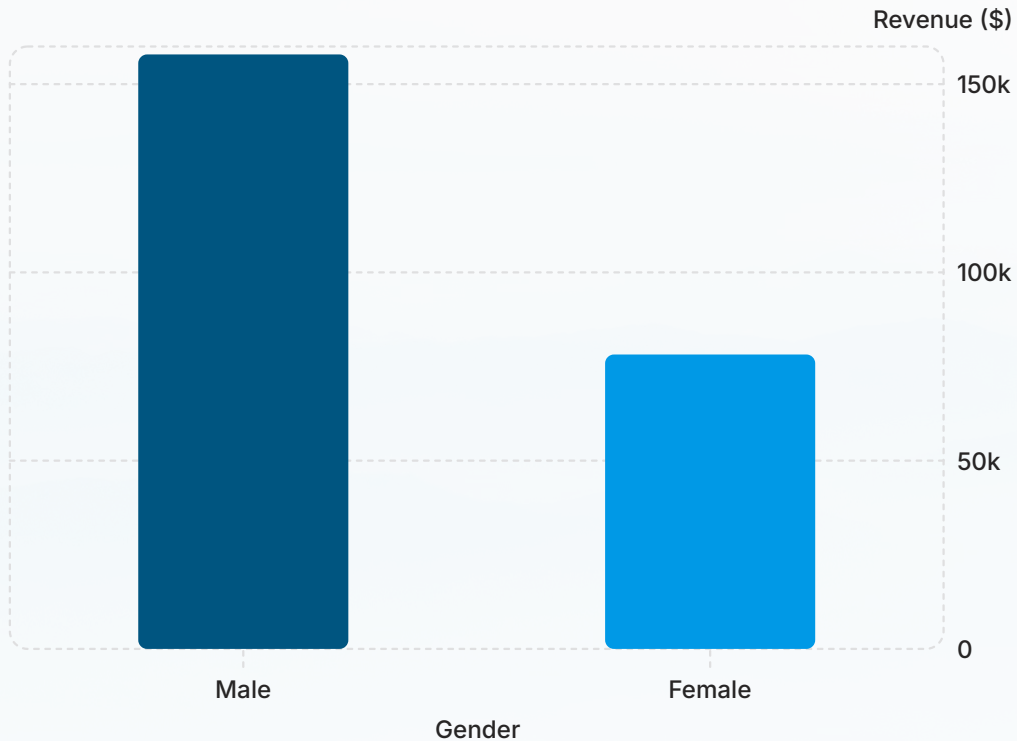
Created `age_group` and `purchase_frequency_days` columns for deeper analysis

06

Database Integration

Connected to MySQL and loaded cleaned DataFrame for SQL analysis

Revenue Insights

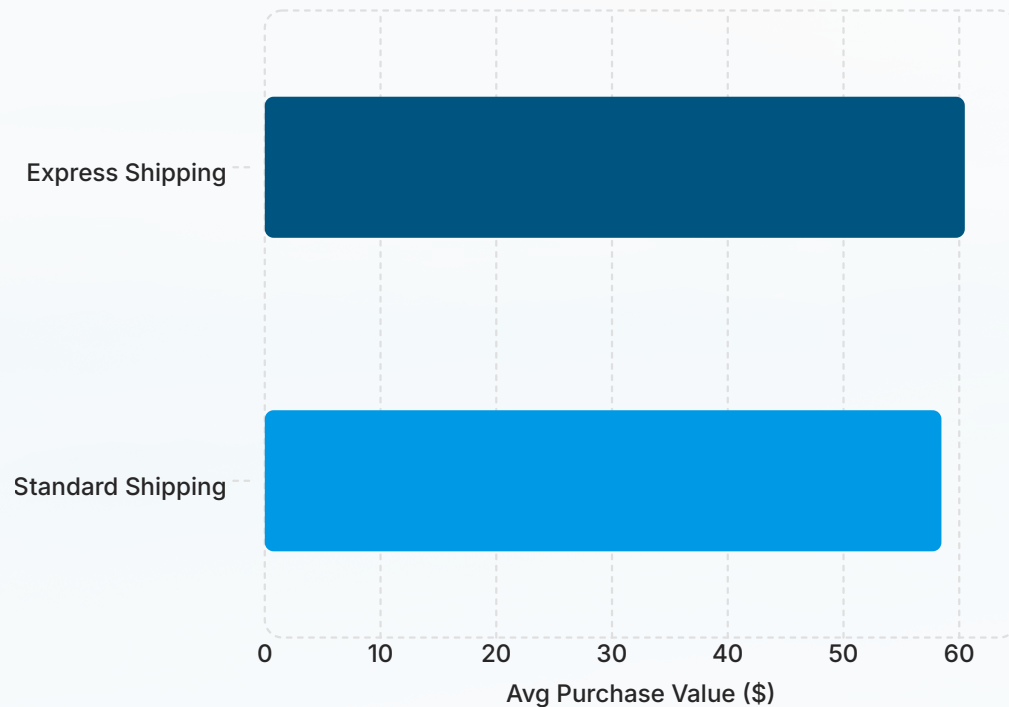


Gender Revenue Distribution

Male customers generated **\$157,890** in total revenue — more than double the **\$78,191** from female customers. This represents a significant demographic gap worth exploring for targeted marketing and product positioning strategies.

Fulfillment & Shipping Impact

Shipping Type



Key Finding

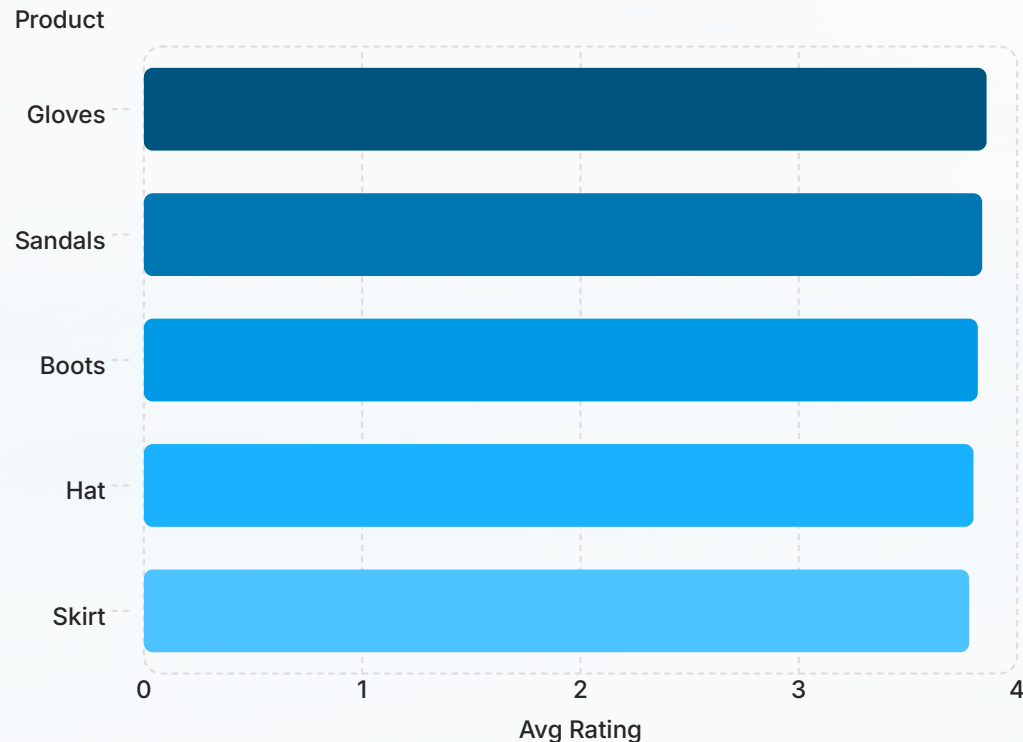
Express shipping cohorts maintain an average purchase value of **\$60.48**, compared to **\$58.46** for Standard delivery.



Operational Takeaway: Higher-value consumer cohorts consistently exhibit a preference for accelerated logistics channels to complete transactions.

Product Performance Analysis

Top 5 Products by Rating



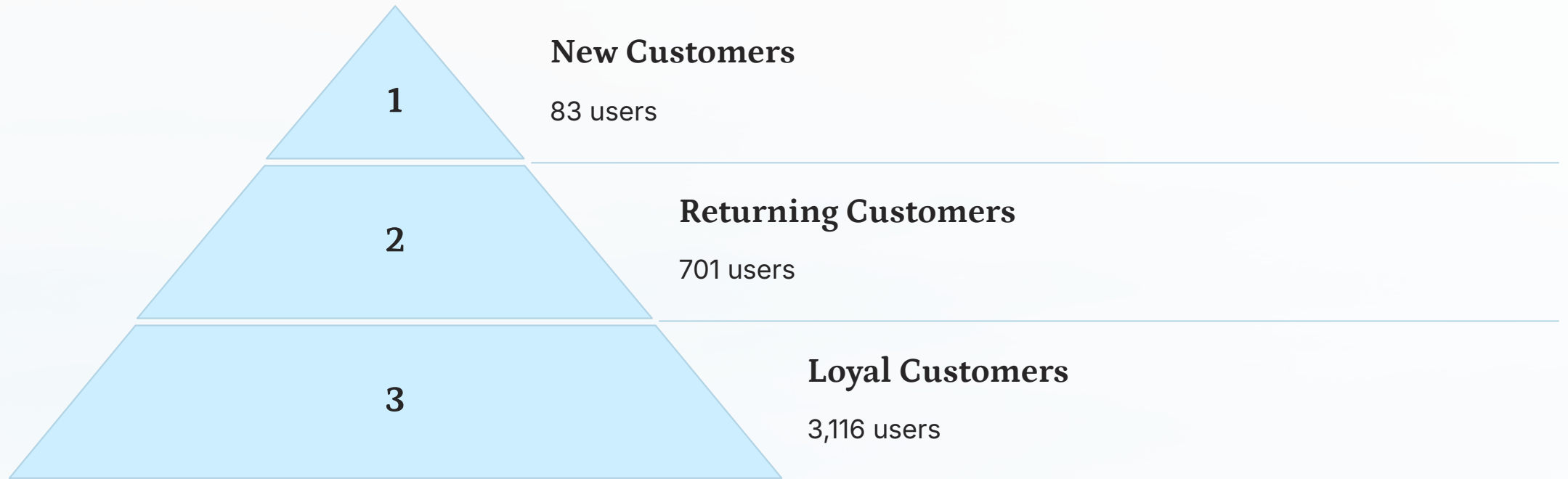
Discount-Dependent Products

Products with the highest percentage of discounted purchases:

- **Hat** — 50% of purchases require a discount
- **Sneakers** — 49.7% discount-dependent
- **Coat** — 49.1% discount-dependent

These categories are highly sensitive to promotional pricing and warrant careful margin management.

Customer Segmentation



Behavioral classification based on historical purchase patterns reveals an exceptionally strong retention architecture, with the **Loyal Customer segment accounting for 80%** of the active customer ecosystem.

Subscription & Top Products by Category

Subscription Value

Active Subscribers

1,053 customers ·
\$59.49 avg spend

Non-Subscribers

2,847 customers ·
\$59.87 avg spend

High-Frequency Buyers

958 subscribers vs 2,518 non-subscribers (purchases >5)

Top Products by Category

- **Accessories:** Jewelry leads with 171 orders; Sunglasses & Belt at 161
- **Clothing:** Blouse & Pants tie at 171; Shirt at 169
- **Footwear:** Sandals top with 160; Shoes 150, Sneakers 145
- **Outerwear:** Jacket leads with 163; Coat at 161

Interactive Power BI Dashboard

We built a comprehensive interactive dashboard in Power BI to visualize key insights and enable data-driven decision making. The interface features a structured navigation sidebar with modules for Regional Analysis, Performance Overview, Predictive Analytics, and Demographic Insights.



Real-Time Insights

Dynamic visualizations of customer behavior and sales trends



Interactive Filters

Drill down by category, demographics, and time periods



KPI Tracking

Monitor key performance indicators —
3.9K active customers, \$59.76 avg
purchase, 3.75 avg rating

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert non-subscribers



Loyalty Programs

Reward repeat buyers to increase lifetime value



Review Discount Policy

Balance sales boosts with margin control on discount-dependent products



Product Positioning

Highlight top-rated and best-selling products in campaigns



Targeted Marketing

Focus on high-revenue age groups and express-shipping users

✅ **Thank You!** We appreciate your time and attention. We're happy to answer any questions regarding our analysis and recommendations.